and in laying out her own cosmic spacetime loaf will assign earth's early bacteria to their appropriate epoch, their appropriate set of time slices.

All of this assumes that both we and the Coma cluster astronomer are moving only with the cosmic flow from spatial expansion, since this ensures that her slicing of the spacetime loaf coincides with ours—it ensures that her *now*-lists agree with ours. However, should she break ranks and move through space substantially in excess of the cosmic flow, her slices will tilt relative to ours, as in Figure 8.9. In this case, as we found with Chewie in Chapter 5, this astronomer's *now* will coincide with what we consider to be our future or our past (depending on whether the additional motion is toward or away from us). Notice, though, that her slices will no longer be spatially homogeneous. Each angled slice in Figure 8.9 intersects the universe in a range of different epochs and so the slices are far from uniform. This significantly complicates the description of cosmic history, which is why physicists and astronomers generally don't contemplate such perspectives. Instead, they usually consider only the perspective of observers moving solely with the cosmic flow, since this yields slices that are homogeneous—but fundamentally speaking, each viewpoint is as valid as any other.

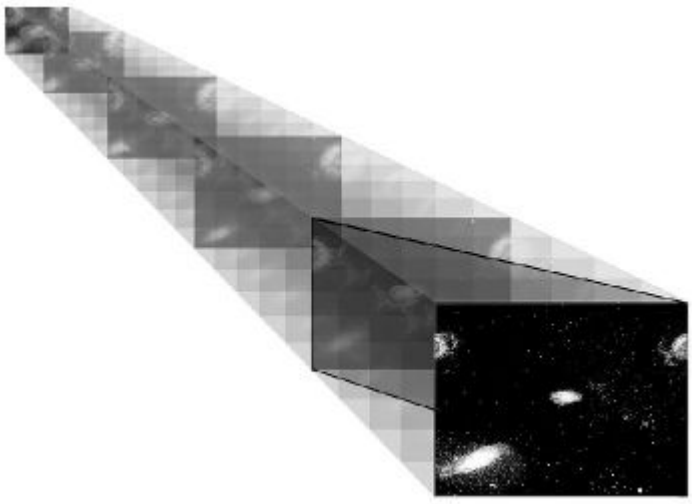


Figure 8.9 The time slice of an observer moving significantly in excess of the cosmic flow from spatial expansion.

As we look farther to the left on the cosmic spacetime loaf, the universe gets ever smaller and ever denser. And just as a bicycle tire gets hotter and hotter as you squeeze more and more air into it, the universe